

GERÓNIMO SALDAÑA ESPINAL

Data Engineer · End-to-End Data Pipelines · Automation & Applied AI

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience across the full data lifecycle — ingestion (web scraping, APIs, documents), processing and cleaning, storage in relational databases, and delivery through APIs and dashboards. Built production data systems including a social-sentiment monitoring platform (FastAPI + PostgreSQL + Celery) and an e-commerce financial analytics pipeline (Go + Python + PostgreSQL). Strong in Python, SQL, and workflow automation, with a track record of turning messy real-world data into reliable, queryable systems.

EXPERIENCE

Implementation Engineer · Secretaría de Planeación, Alcaldía de Tuluá 2024

Direct municipal contract · Embeddings · OCR · Semantic Search · PostgreSQL

- Built a document-intelligence pipeline that indexed 3,000 OCR-scanned documents into a vector-embeddings store, replacing physical folder archives with instant semantic search.
- Owned the full path from data ingestion and processing to end-user adoption: trained 7 staff members and supported sustained daily usage by institutional users.

Data & Automation Engineer · Senate Campaign of Juan Camilo Vélez 2024 – 2025

Employment · n8n · Chatwoot · PostgreSQL · APIs

- Automated 221 of 300 inbound conversations end-to-end (~74%, tracked via Chatwoot tags) through n8n workflows integrating AI agents and human-agent handoff.
- Designed an automated ingestion-to-storage flow logging 600+ inbound messages into structured datasets (name, message, phone), giving the team a real-time queryable record without manual transcription.

SELECTED DATA ENGINEERING PROJECTS

Termómetro Cultural · Social-Sentiment Monitoring Platform 2025

Python · FastAPI · PostgreSQL · Celery · Redis · SQLAlchemy · LLMs

- Architected a full ingestion-to-API data system: scheduled scrapers (Celery Beat) ingest social posts and news, an NLP pipeline classifies sentiment, topic, and urgency via LLMs, results persist to PostgreSQL, and a FastAPI layer exposes analytics endpoints.
- Designed the relational schema (sources, posts, analysis_results, topics, sentiment) with cached aggregation columns for fast querying, and built Alembic migrations and a Dockerized multi-service deployment (API, DB, Redis, worker, beat).
- Implemented automated PII sanitization (Colombian Data Protection Law 1581) in the processing layer and webhook integrations for downstream consumers.

MasterMetrics · E-commerce Financial Data Pipeline 2025 – in progress

Go · Python · PostgreSQL · SQL · Browser Automation

- Building a hybrid pipeline (Go ingestion backend + SQL business rules) that automates extraction of e-commerce sales reports, deduplicates by row hash, assigns a unified order ID, and reconciles transactional and wallet data.
- Designed the core data model (unified_orders, order_items, wallet_transactions, unified_customers) with full load auditing and incremental, re-runnable processing, replacing fragile Excel macros with automated KPI computation (P&L, return rate, average ticket, recurrence).

AI Workflow Automation & Self-Hosted Infrastructure 2024 – Present

Personal / Freelance · n8n · LangChain · Python · Docker · REST APIs

- Built a lead-management automation pipeline integrating webhooks, AI agents, and a CRM via REST APIs (validating ~40 weekly leads). Reduced manual operating time from 4 to 1.5 hours/week (measured across 3 consecutive weeks before/after).
- Operates self-hosted infrastructure (CasaOS + Docker) running 10+ containerized microservices for personal data and automation services.

Predictive Analytics & Customer Segmentation · Construction Sector Client 2023

Freelance · Python · Scikit-learn · Power BI

- Built customer segmentation (K-Means, DBSCAN) and a behavioral targeting model for paid campaigns; CPA decreased from \$120 to \$85 across 3 consecutive months on comparable budgets, with the model validated offline at +12% AUC over baseline. Delivered Power BI dashboards with automated ETL.

ACADEMIC BIG DATA PROJECT

Climate Data Analysis (CRISP-DM) · University of Manizales 2023

PySpark · SparkML · IBM Cloud · Python

- Applied the full CRISP-DM methodology on multi-year climate datasets (>500k records) using distributed processing (Apache Spark on a multi-node cluster), with feature engineering and model evaluation against a baseline.

TECHNICAL SKILLS

Languages & Databases: Python (Advanced), SQL / PostgreSQL / MySQL (Advanced), Go, PHP

Data Engineering: ETL/ELT pipelines, SQLAlchemy, Alembic, Celery, Redis, FastAPI, REST APIs, Webhooks, web scraping (Playwright, BeautifulSoup)

Big Data: Apache Spark / PySpark, SparkSQL, Hadoop / HDFS, Apache Airflow, Databricks

Cloud & Infrastructure: Microsoft Azure (Data Factory), IBM Cloud, Docker, Git / GitHub (note: AWS not yet hands-on — concepts transferable)

AI & Automation: n8n, LangChain, AI Agents, OpenAI / Grok APIs, Prompt Engineering

Analytics & Visualization: Power BI (Advanced), Pandas, NumPy, Scikit-learn, Apache ECharts

Methodologies: SCRUM, CRISP-DM, Medallion Architecture, Software Quality Models

EDUCATION

B.S. in Data Analytics Engineering · University of Manizales 2021 – 2026 (Expected)

MeMo & BIEN Scholarship recipient (academic merit, 4+ consecutive semesters).

Associate Degree in Computer Programming · SENA 2020 – 2021

CERTIFICATIONS (SELECTED)

IBM / edX: Apache Spark for Data Engineering & ML · Big Data, Hadoop & Spark · NoSQL, Big Data & Spark Fundamentals

Databricks: AI Agents Fundamentals · Generative AI Fundamentals (2026) | **Platzi:** Docker · Flask Backend · PyTorch · Software Engineering & SQL

UPValencia / edX: Prompt Engineering & Productivity | **SENA:** Software Quality Models · Relational Database Design

LANGUAGES

Spanish (Native) · English (B2 — Professional Working Proficiency) · Chinese (Basic)